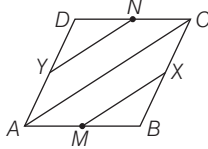


Now, in $\triangle ACE$, $x^2 + a^2 = AC^2$
 $\Rightarrow AC^2 = b^2 + a^2$
 $\therefore AC = \sqrt{b^2 + a^2}$

60. (b) Given, $ABCD$ is a parallelogram. X and Y are mid-points of BC and AD , respectively. M and N are the mid-points of AB and CD , respectively.



Now, join AC .

In $\triangle ABC$, M and X are mid-points of AB and BC .

$$\therefore MX \parallel AC \text{ and } MX = \frac{1}{2} AC \quad \dots(i)$$

In $\triangle ADC$, Y and N are mid-points of AD and CD .

$$\therefore YN \parallel AC \text{ and } YN = \frac{1}{2} AC \quad \dots(ii)$$

From Eqs. (i) and (ii), we get

$$MX \parallel YN$$

So, statement I is not correct.

II Obviously, straight lines AC, BD, XY and MN meet at a point. So, statement II is correct.

61. (c) Given, each interior angle of a regular polygon = 135°

$\therefore$ Each exterior angle

$$= 180^\circ - \text{interior angle} \\ = 180^\circ - 135^\circ = 45^\circ$$

$$\therefore \text{Number of sides in the polygon } (n) \\ = \frac{360^\circ}{\text{Each exterior angle}} = \frac{360^\circ}{45^\circ} = 8$$

$$\therefore \text{Number of diagonals of a polygon} \\ = \frac{n(n-1)}{2} - n = \frac{8(8-1)}{2} - 8 \\ = \frac{8 \times 7}{2} - 8 \\ = 28 - 8 = 20$$

Hence, the number of diagonals of a polygon is 20.

62. (a) $\therefore$ Each interior angle of a regular polygon of n side

$$= \frac{(n-2) \times 180^\circ}{n}$$

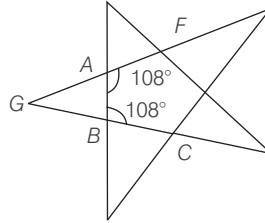
Here, $n = 5$

$\therefore$ Each interior angle of the pentagon

$$= \frac{(5-2) \times 180^\circ}{5} = \frac{3 \times 180^\circ}{5}$$

$$= 3 \times 36^\circ = 108^\circ$$

Since, GAF is a straight line.



$$\therefore \angle GAF = 180^\circ$$

$$\Rightarrow \angle GAB + \angle BAF = 180^\circ$$

$$\Rightarrow \angle GAB + 108^\circ = 180^\circ$$

$$\Rightarrow \angle GAB = 180^\circ - 108^\circ = 72^\circ$$

Similarly,

$$\angle GBA = 180^\circ - 108^\circ = 72^\circ$$

In $\triangle ABG$,

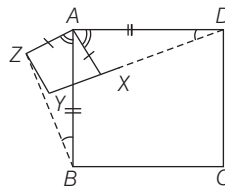
$$\angle GAB + \angle ABG + \angle BGA = 180^\circ$$

$$\Rightarrow 72^\circ + 72^\circ + \angle BGA = 180^\circ$$

$$\Rightarrow \angle BGA = 180^\circ - 144^\circ, \angle BGA = 36^\circ$$

63. (c) Here, $\angle DAB = \angle XAZ = 90^\circ$

$$\Rightarrow \angle DAB - \angle XAB = \angle XAZ - \angle XAB$$



$$\Rightarrow \angle DAX = \angle BAZ$$

Now, in $\triangle DAX$ and $\triangle BAZ$,

$$AD = AB \quad (\text{sides of squares})$$

$$\Rightarrow \angle DAX = \angle BAZ \quad (\text{proved earlier})$$

$$\Rightarrow AX = AZ \quad (\text{sides of square})$$

$$\Rightarrow \triangle DAX \cong \triangle BAZ \quad (\text{SAS criterion})$$

$$\Rightarrow BZ = DX \quad (\text{by CPCT})$$

$$\text{and } \angle ABZ = \angle ADX \quad (\text{by CPCT})$$

Hence, both statements are correct.

64. (d) Given as $\frac{PB}{PC} = \frac{1}{2} \quad \dots(i)$

$$BQ \parallel DC \quad (\because AB \parallel DC)$$

In $\triangle BPQ$ and $\triangle DPC$

$$\angle BPQ = \angle DPC$$

($\because$ vertically opposite angles)

$$\Rightarrow \angle QBP = \angle PCD$$

($\because$ alternate interior angle)

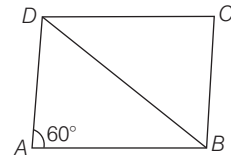
$$\Rightarrow \triangle BPQ \sim \triangle DPC \quad [\text{by AA similarity}]$$

We know that area of two similar triangles are in the ratio of the square of the corresponding sides

$$\frac{\text{ar}(\triangle BPQ)}{\text{ar}(\triangle DPC)} = \left(\frac{PB}{PC}\right)^2 \\ \Rightarrow \frac{20}{\text{ar}(\triangle DPC)} = \left(\frac{1}{2}\right)^2 \quad [\text{from Eq. (i)}]$$

$$\therefore \text{ar}(\triangle DPC) = 80 \text{ sq units}$$

65. (b)



Let $AD = x$, then $AB = 2x$

using cosine rule in $\triangle ABD$, we have

$$\cos A = \frac{AD^2 + AB^2 - BD^2}{2 \cdot AD \cdot AB} \\ \Rightarrow \cos 60^\circ = \frac{x^2 + (2x)^2 - BD^2}{2 \cdot x \cdot (2x)}$$

$$\Rightarrow \frac{1}{2} = \frac{5x^2 - BD^2}{4x^2}$$

$$\Rightarrow 2x^2 = 5x^2 - BD^2$$

$$\Rightarrow BD^2 = 3x^2$$

$$\Rightarrow BD = \sqrt{3}x = \sqrt{3}AD.$$

66. (b) Given, each interior angle = 140°

Then, each exterior angle

$$= (180^\circ - 140^\circ) = 40^\circ$$

$\therefore$ Number of sides

$$= \frac{360^\circ}{\text{Each exterior angle}} = \frac{360^\circ}{40^\circ} = 9$$

Hence, the number of vertices of polygon is 9.

67. (b) Exterior angle of regular polygon

$$= \frac{360^\circ}{\text{Number of sides}}$$

I. If exterior angle is 70°

Then, number of sides

$$= \frac{360^\circ}{70^\circ} = \frac{36}{7} = \text{A rational number}$$

So, statement I is incorrect.

II. If number of sides ≥ 5 , then

$$\text{Exterior angle} = \frac{360^\circ}{5} = 72^\circ$$

For number of sides ≥ 5 , there is always an acute exterior angle.

So, statement II is correct.

68. (d) I. In regular polygon,

$$\text{Sum of exterior angles} = 360^\circ$$

[which is always constant]

So, statements I is incorrect.

II. Sum of interior angles of m sides of regular polygon = $(m-2) \times 180^\circ$

Sum of interior angles of m sides and

n sides of regular polygon

$$= (m-2) \times 180^\circ + (n-2) \times 180^\circ$$

$$= (m+n-4) \times 180^\circ$$

So, statement II is incorrect.

Hence, neither I nor II is correct.

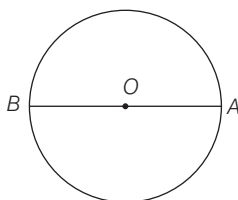
CIRCLE

Usually (3-6) questions have been asked from this chapter. Generally questions are asked from the topics related to theorem of circle, tangent to a circle and locus.



A circle is a curved figure consisting of all points in a plane that are at a fixed distance from a fixed point. The fixed point is called centre and the fixed distance is called radius of the circle.

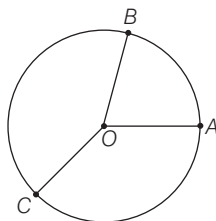
- Circles are simple closed curves which divide the plane into two regions : an interior and an exterior.
- In the given figure, O is the centre of circle, OA is the radius of the circle and AB is the diameter of the circle.



RADIUS

Radius is the fixed distance between the centre of the circle and the points lying on the circle.

Here, in the given figure OA , OB and OC all are having fixed distance, so they are known as radius of the circle. It is denoted by r .

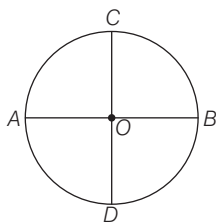


Diameter

Diameter is any line segment that passes through the centre of the circle and whose end points lie on the circle. The diameter of a circle is twice the radius. Here, AB and CD are the diameters of the circle.

$$AB = CD = 2(OA) \\ = 2(OB) = 2(OC) = 2(OD)$$

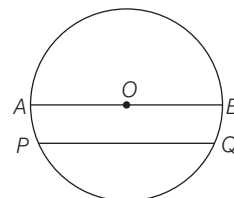
A circle can have an infinite number of diameters.



Chord

A chord is a line segment whose end points lie on the circle. A diameter is the longest chord in a circle.

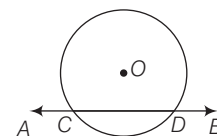
Here, in the given figure PQ is a chord and AB is the diameter of the circle which is the longest chord of that circle.



Secant

A line which intersects the circle at two points is called secant of the circle.

Here, in the figure AB is a secant cutting the circle at two distinct points as C and D .

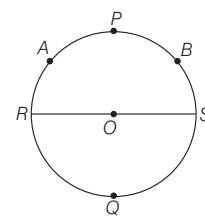


Arc

Any part of a circle between two points is called an arc of the circle.

Any two points, say A and B of a circle divide it into two parts called arcs. The smaller arc APB is called minor arc and larger arc AQB is called major arc.

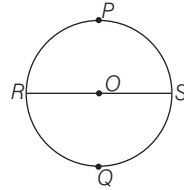
It is usually denoted by $\widehat{APB}$ and $\widehat{AQB}$.



Semi Circle

A diameter divides the circle into two equal parts. Each of the two is called a semi-circle.

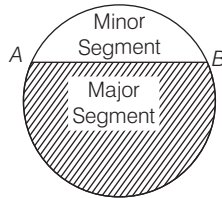
In the above figure, $RPSR$ and $RQSR$ are semi-circles.



Segment

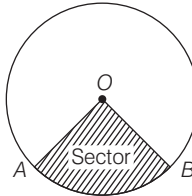
The area enclosed by an arc and its corresponding chord is called a segment of the circle.

- The segment containing the minor arc is called minor segment.
- The segment containing the major arc is called major segment.



Sector

The area enclosed by any two radii and the arc determined by the end points of the radii is called a sector of the circle.



Circumference

The length of the complete circle is called its circumference.

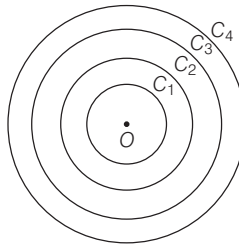
$$\begin{aligned} \text{Circumference} &= 2 \times \pi \times \text{Radius of the circle} \\ &\text{or} \\ &= \pi \times \text{Diameter of the circle} \end{aligned}$$

Concentric Circles

Two or more circles having the same centre are called concentric circles.

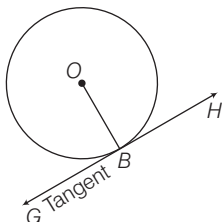
Here, in the given figure C_1, C_2, C_3 and C_4 are known as concentric circles as they have a common centre 'O'.

An infinite number of circles can be drawn with same centre.



Tangent

A tangent to a circle is a straight line that touches the circle at a single point.

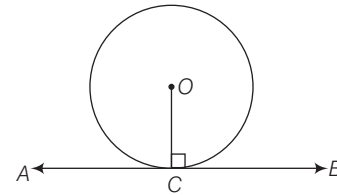


Here, in the given figure GBH is a tangent touching the circle at a single point B .

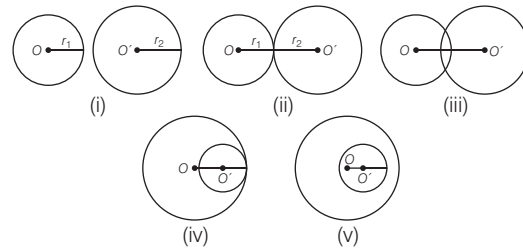
Length of tangent The distance between external point P from which tangent is drawn and the point of contact of the tangent is called length of tangent.

Properties of Tangent To a Circle

- Only two tangents can be drawn from a point outside the circle.
- Only one tangent can be drawn through a point lying on the circle.
- No tangent can be drawn from a point lying inside the circle.
- A tangent at any point of a circle is perpendicular to the radius through the point of contact.



Relative Position of two Circles



Two circles will

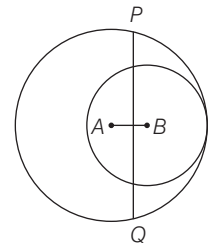
- be disjoint, when $OO' > r_1 + r_2$.
- be touching externally, when $OO' = r_1 + r_2$.
- be intersecting when $OO' < r_1 + r_2$.
- be touching internally, when $OO' = |r_2 - r_1|$
- one of the circle will lie inside the other, when $OO' < |r_2 - r_1|$.

EXAMPLE 1. Two circles with centres A and B touch each other internally, as shown in the figure given below. Their radii are 5 and 3 units, respectively. Perpendicular bisector of AB meets the bigger circle in P and Q . What is the length of PQ ?

- a. $2\sqrt{6}$ b. $\sqrt{34}$ c. $4\sqrt{6}$ d. $6\sqrt{2}$

Sol. c. Let PQ intersect AB at O . Join AP and AQ . Produce AB to intersect the circle at M .

Here, $AP = AM = 5$ units, $BM = 3$ units



$$\therefore AB = AM - BM = 5 - 3 = 2 \text{ units}$$

PQ bisects AB perpendicularly.

$$\therefore AO = OB = 1 \text{ unit}$$

Now, in right $\triangle AOP$,

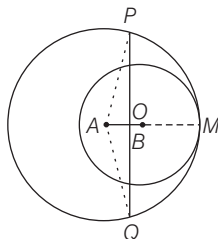
$$(PA)^2 = (OA)^2 + (OP)^2$$

$$\Rightarrow (5)^2 = (1)^2 + (OP)^2$$

$$\Rightarrow 24 = (OP)^2 \Rightarrow OP = 2\sqrt{6} \text{ units}$$

$$\therefore PQ = 2OP = 2 \times 2\sqrt{6} = 4\sqrt{6} \text{ units}$$

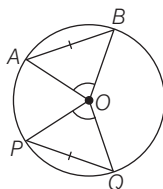
Hence, the length of PQ is $4\sqrt{6}$ units.



Important Theorems of Circles

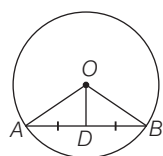
Theorem 1 If two arcs of a circle are congruent, then the corresponding chords are equal.

Theorem 2 Equal arcs (or chords) subtend equal angle at the centre. Here, if $AB = PQ$, then $\angle AOB = \angle POQ$.



Theorem 3 The perpendicular from the centre of a circle to a chord bisects the chord.

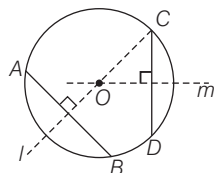
Here, if $OD \perp AB$, then $AD = DB$.



Theorem 4 The line joining the centre to the mid-point of a chord is perpendicular to the chord.

Here, if $AD = DB$, then $\angle ADO = \angle ODB = 90^\circ$.

Theorem 5 The perpendicular bisectors of two chords of a circle intersect at its centre.

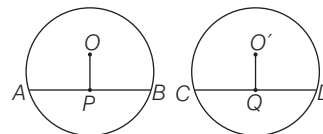


Here, AB, CD are the chords and l, m are perpendicular bisector of AB and CD . So, l and m meet at centre 'O'.

Theorem 6 There is one and only one circle passing through three non-collinear points.

- An infinite number of circles can be drawn to pass through a single point.
- An infinite number of circles can be drawn to pass through two given points.
- A unique circle can be drawn to pass through three given non-collinear points.

Theorem 7 Equal chords of congruent circles are equidistant from the corresponding centres.

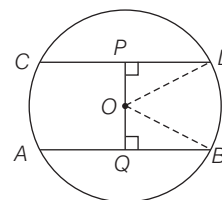


Here, if $AB = CD$, then $OP = O'Q$.

Conversely if $OP = O'Q$, then $AB = CD$.

Theorem 8 Equal chords of a circle are equidistant from the centre.

Here, if AB and CD are equal chords of circle, then $OP = OQ$.



Conversely, if $OP = OQ$, then also $AB = CD$

i.e. chords at equal distance from the centre are equal.

EXAMPLE 2. AB and CD are two parallel chords on the opposite sides of the centre of the circle. If $AB = 10$ cm, $CD = 24$ cm and the radius of the circle is 13 cm, then what is the distance between the chords?

- a. 10 cm b. 17 cm
c. 24 cm d. None of these

Sol. b. From O draw $OL \perp AB$ and $OM \perp CD$. Join OA and OC .

$$AL = \frac{1}{2} AB = \frac{10}{2} \text{ cm} = 5 \text{ cm},$$

$$OA = 13 \text{ cm}$$

In right angled $\triangle OLA$, by pythagoras theorem,

$$OL^2 = OA^2 - AL^2 = (13)^2 - (5)^2$$

$$= (169 - 25) = 144$$

$$\Rightarrow OL = \sqrt{144} = 12 \text{ cm}$$

$$\text{Now, } CM = \frac{1}{2} \times CD = 12 \text{ cm}$$

$$\text{and } OC = 13 \text{ cm}$$

In $\triangle OMC$,

$$OM^2 = OC^2 - CM^2$$

$$= (13)^2 - (12)^2$$

$$= (169 - 144) = 25$$

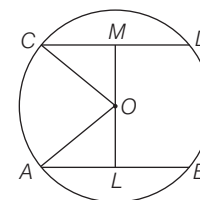
$$\Rightarrow OM = \sqrt{25} = 5 \text{ cm}$$

$$\therefore ML = OM + OL$$

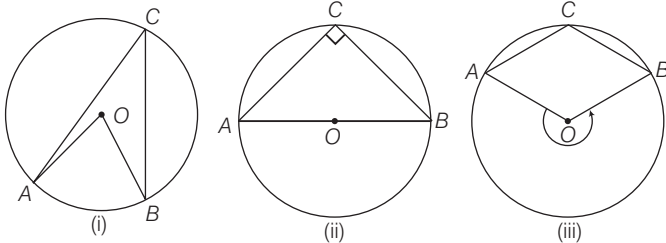
$$= (5 + 12) \text{ cm}$$

$$= 17 \text{ cm}$$

[by pythagoras theorem]

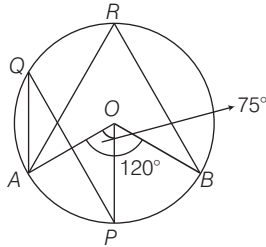


Theorem 9 The angle subtended by an arc of a circle at the centre is double the angle subtended by it at any point on the circumference of circle. Here, three case arises.



In all the three cases $\angle AOB = 2\angle ACB$.

EXAMPLE 3 In the figure given below, if $\angle AOP = 75^\circ$ and $\angle AOB = 120^\circ$, then what is the value of $\angle AQP$?



- a. 45° b. 37.5° c. 30° d. 22.5°

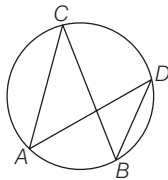
Sol. b. We know that the angle subtended by an arc of a circle at the centre is double the angle subtended by it at any point on the circumference of a circle.

$$2\angle AQP = \angle AOP$$

$$\angle AQP = \frac{1}{2} \times \angle AOP = \frac{75^\circ}{2} = 37.5^\circ$$

Theorem 10 The angle in a semi-circle is a right angle.

Theorem 11 Angles in the same segment of a circle are equal.

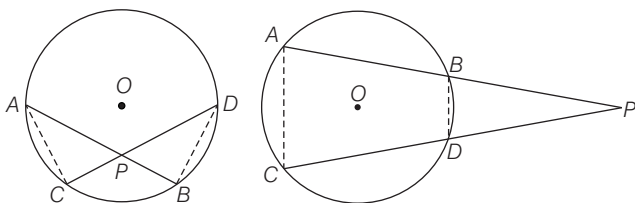


Here, $\angle ACB = \angle ADB$

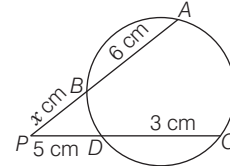
Theorem 12 If two chords AB and CD of a circle intersect inside or outside the circle when produced at a point P

then,

$$AP \times PB = DP \times PC$$



EXAMPLE 4. In the given figure, chords AB and CD of a circle intersect externally at P . If $AB = 6$ cm, $CD = 3$ cm and $PD = 5$ cm, then the measurement of PB is



- a. 2.5 cm b. 4 cm c. 10 cm d. None of these

Sol. b. Let the measurement of PB be x .

$$PA \times PB = PC \times PD \Rightarrow (x + 6) \times x = 8 \times 5$$

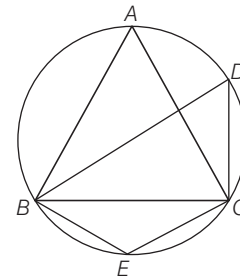
$$\Rightarrow x^2 + 6x - 40 = 0 \Rightarrow x^2 + 10x - 4x - 40 = 0$$

$$\Rightarrow x(x + 10) - 4(x + 10) = 0 \Rightarrow (x + 10)(x - 4) = 0$$

$$\Rightarrow x = 4, \text{ as } x \neq -10 \quad [\text{side cannot be negative}]$$

$$\therefore PB = 4 \text{ cm}$$

EXAMPLE 5. In the adjoining figure, $\triangle ABC$ is an isosceles triangle with $AB = AC$ and $\angle ABC = 50^\circ$. Then, the measure of $\angle BDC$ is



- a. 80° b. 100° c. 40° d. 160°

Sol. a. Given, $AB = AC \Rightarrow \angle ACB = \angle ABC = 50^\circ$

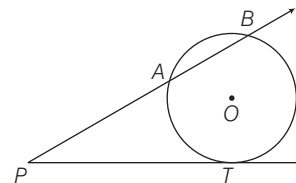
$$\text{In } \triangle ABC, \angle BAC = 180^\circ - (50^\circ + 50^\circ) = 80^\circ$$

[by angle sum property of a triangle]

$$\therefore \angle BDC = \angle BAC = 80^\circ$$

[angles lying in the same segment of the circle]

Theorem 13 Let PT be a line tangent to the circle at point T . From external point P , draw a secant which intersect the circle at two point A and B , then $PA \times PB = PT^2$



EXAMPLE 6. In a $\triangle ABC$, $AB = AC$. A circle through B touches AC at D and intersects AB at P . If D is the mid-point of AC , then which one of the following is correct?

- a. $AB = 2AP$ b. $AB = 3AP$ c. $AB = 4AP$ d. $2AB = 5AP$